

Would you trust this preview?

It looks like a real photo.
But the shirt has changed color.

How closely can AI reproduce
the clothes we choose?



Requested garment



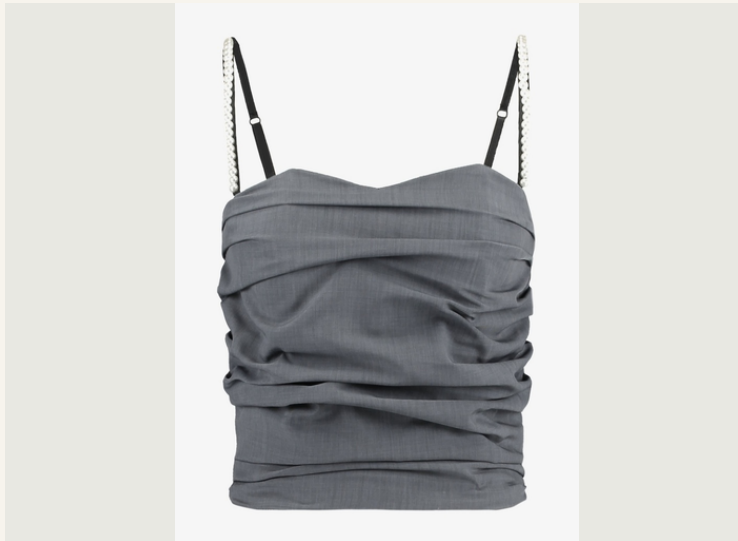
Pretrained preview

Selected example: 07703_00, seed 9026. One example, not a failure-rate estimate.

Better garment reconstruction, without unwanted changes elsewhere?



Edited person input



Requested garment



Original reference

1,000 training cases

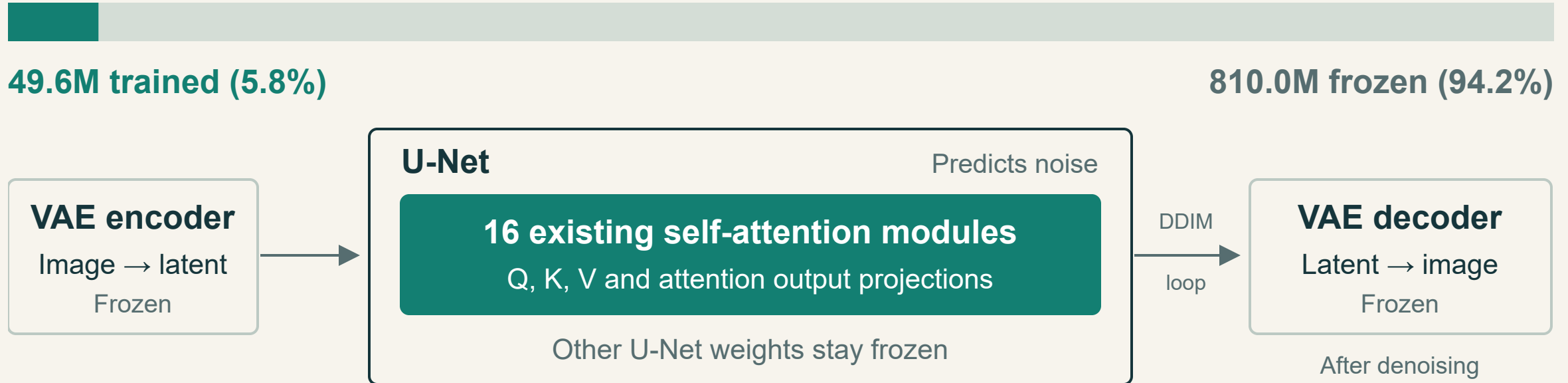
64 validation cases

32 final-test cases

VITON-HD + VITON-HD-edit. Custom reconstruction study for clothing-preview developers.

859.5 million parameters. I updated 5.8%.

U-Net parameters in the recorded run. The VAE is separate and frozen. Starting point: CatVTON-MaskFree.



Training signal

Predict the noise added to the target latent.

Noise MSE updates attention only.

Controlled adaptation

1,000 cases · 3 epochs

3,000 optimizer updates

AdamW · learning rate 0.00001

Verified checkpoint

Frozen weights unchanged

Saved weights reload exactly

Checked against recorded tensor hashes.

A paired test of reconstruction and preservation

Same person, garment, resolution, generation steps and random seeds for both models.



Clothing region

MAE + SSIM against the original reference

Outside clothing

MAE against the input person photograph

Two seeds per model → Average within case → Paired difference

The case is the analysis unit. Two outputs of the same case are not two independent cases.

Masks define the measurement regions; no mask is supplied to the generator.

32 × 2 models × 2 seeds

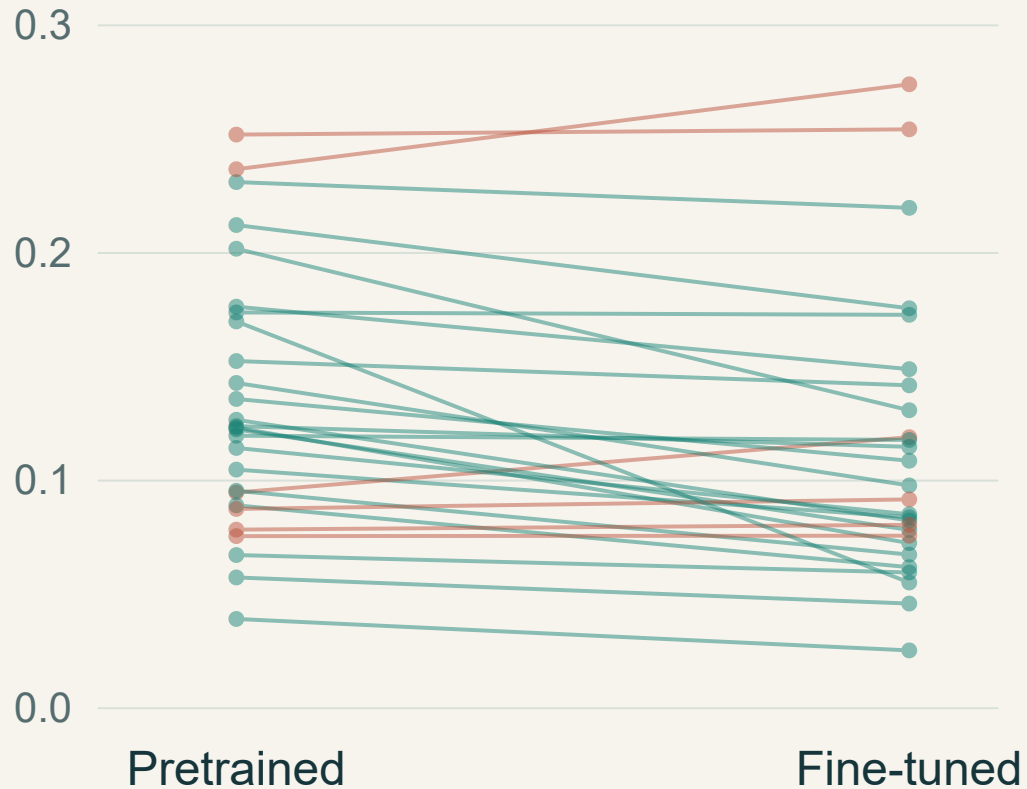
128 attempts → **27 complete pairs**

Six safety exclusions across five cases; no retries.

384 × 512 · 20 DDIM steps · guidance 2.5

10,000 paired case-bootstrap resamples for uncertainty.

Garment error fell 15.6% on average



Each line is one case. Lower error is better.

21 / 27

cases improved

Mean MAE

0.1335 pretrained

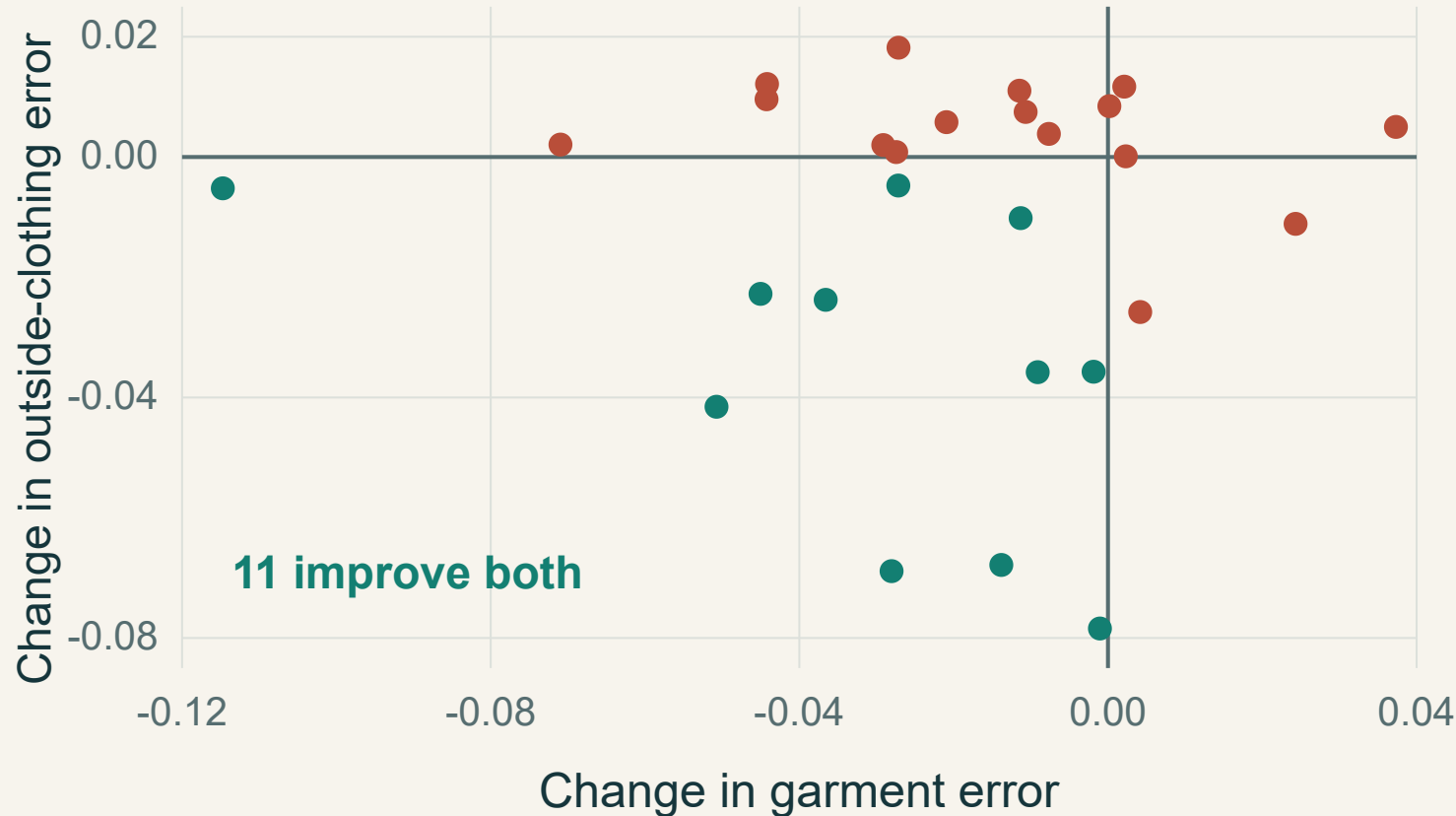
0.1127 fine-tuned

Green lines improve.

Six red lines worsen.

Paired mean difference: -0.02085 . Descriptive 95% case-bootstrap interval: $[-0.03222, -0.01032]$.

Preservation did not improve for every person



14 / 27

cases had more
outside-clothing error

Yet average outside error
fell 25.9%.

Larger gains in some cases
outweighed smaller regressions.

Each dot is one paired case. Differences are fine-tuned minus pretrained. Lower left improves both errors.

Even the largest gain still changes the person



Input person



Pretrained



Fine-tuned



Reference

The clothing is closer to the reference.

The arm pose and trousers still change.

Case 08931_00, seed 9026. Selected by the largest case-average garment-MAE reduction. Full gallery includes all cases.

A closer clothing match. Still unwanted changes elsewhere.

What the evidence supports

15.6% lower mean garment error.
Only 11 of 27 cases improved
both clothing and preservation.

My recommendation

Keep Klara as a research prototype.
Test independent, more varied data
and repeat training before wider use.

My contribution

Data verification → attention adaptation → paired evaluation → deployed demo

Limits: edited inputs, 27 complete pairs, one training seed.
Identity separation and upstream exposure remain unverified.
These images do not establish physical fit.

klara-app.de

github.com/anasm87/klara_tryon

The training target is known noise

P = input person · G = requested garment · T = original target. Frozen VAE encodes each to [B, 4, 64, 48].

1. Clean target canvas

Target T	Garment G
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$z_0 = \text{cat_width}(T, G) \quad [B, 4, 64, 96]$

3. Sample noise and a timestep

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{(1 - \bar{\alpha}_t)} \varepsilon$$

$\varepsilon \sim N(0, I)$ · fresh noise and t at every update.
 $\bar{\alpha}_t$ is the scheduler's cumulative signal retention.

2. Fixed conditioning

Person P	Garment G
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$C = \text{cat_width}(P, G) \quad [B, 4, 64, 96]$

4. Predict that noise

$$\hat{\varepsilon} = \text{U-Net}_{\theta}(\text{cat_channels}(z_t, C), t)$$

Input [B, 8, 64, 96] → output [B, 4, 64, 96]
 One U-Net pass per training example; batch size 1.

5. Optimize attention weights only

$$L = \text{mean}[(\hat{\varepsilon} - \varepsilon)^2]$$

backward() → clip gradients → AdamW.step()

Both canvas halves contribute to MSE. No regional penalty in this run.

Training: code/training-feasibility/training_core.py and evidence/training-1000-01/run.json.

Garment conditioning is zeroed with probability 0.1; the clean target is unchanged. No 50-step denoising loop during training.

Exactly which attention weights changed

Example: the first U-Net attention stage at research resolution. Eight heads, each with dimension 40.

Feature map [B, 320, 64, 96]

GroupNorm → outer projection → tokens X [B, 6144, 320]

$$\mathbf{Q} = \text{LN}(\mathbf{X})\mathbf{W}_{\mathbf{Q}} \quad \mathbf{K} = \text{LN}(\mathbf{X})\mathbf{W}_{\mathbf{K}} \quad \mathbf{V} = \text{LN}(\mathbf{X})\mathbf{W}_{\mathbf{V}}$$

Learned projections; split each into [B, 8, 6144, 40]

$$\mathbf{A} = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{40}) \rightarrow \mathbf{AV}$$

Conceptual scores [B, 8, 6144, 6144]; no causal mask

$$\mathbf{Y} = \mathbf{X} + \text{concat_heads}(\mathbf{AV})\mathbf{W}_{\text{out}} + \mathbf{b}_{\text{out}}$$

Attention output projection mixes the joined heads.

49.6M

of 859.5M U-Net
parameters updated

Training scope

Q / K / V / output weights
inside 16 attn1 modules.

The outer Transformer projections, feed-forward layers, norms, convolutions and VAE remain frozen.

Green marks learned projections. Softmax and residual addition have no trainable parameters.

Attention connects person and garment features; exact colors, identity and physical fit are not guaranteed.

Source: training_core.py, pinned vendor architecture and run.json. Remaining Transformer operations omitted from this attn1 detail.

What makes the result credible, and limited

Data and model controls

1,000 / 64 / 32 cases

Source groups and exact duplicates checked

Pinned dataset and model revisions

Custom split of the original VITON-HD test partition.

Identity separation and upstream exposure unverified.

Regional measurement

Clothing: MAE / SSIM vs target

Outside: MAE vs input person

Conservative mask boundaries

Clothing: erode 3 px; exclude a 3 px image border.

Outside: exclude 8 px dilation of source/target mask union.

Comparison and uncertainty

Matched inputs, settings and seeds

Seed means within each case

Bootstrap 27 paired case differences

Δ garment MAE = -0.02085

Descriptive 95% interval $[-0.03222, -0.01032]$.

Limits that affect interpretation

Six exclusions without retries

Edited photos; one training seed

No direct test of identity, preference or fit

Complete-pair exclusions may bias the 27 remaining cases.

Outside pixel error is a proxy for unwanted visual changes.

Reproducibility: code, pinned inputs, run logs, checkpoint hash and per-case outputs

Research: 384×512 , 20 steps, guidance 2.5, seeds 9026 / 9027. Website: 768×1024 , 50 steps.